

Bitcoin Price Prediction

AI Dashboard

D16,093.010



BitPredict

Deep Learning Bitcoin Time Series Forecasting

AI-Powered Cryptocurrency Price Forecasting using TensorFlow

AI ENGINEERING

TENSORFLOW DEEP LEARNING

FINANCIAL AI

Project Overview



Objective

Forecast future Bitcoin prices using historical market data through multiple deep learning architectures — built for research-grade experimentation and production-ready accuracy.

Time Series Forecasting

Sequential pattern learning across market history

Financial Data Analysis

Multi-feature market signal processing

Deep Learning Architectures

LSTM, Conv1D, N-BEATS, and ensemble models

Dataset & Data Engineering



Engineering Rigor

Every stage of the pipeline is designed to preserve temporal integrity and maximize signal quality for deep learning models.

Source

Historical BTC/USD daily OHLCV

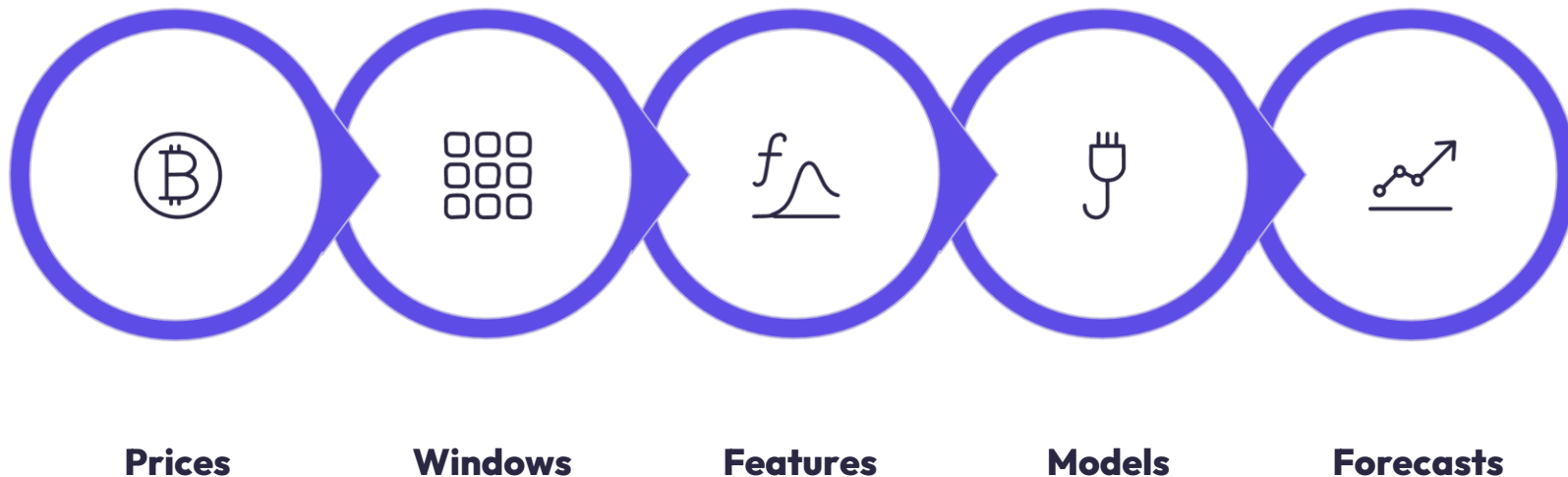
Split

Chronological — no leakage

Windows

Sliding sequence generation

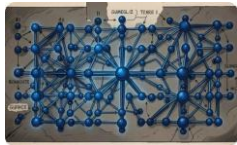
End-to-End Forecasting Pipeline



A fully automated pipeline transforms raw market data into calibrated price forecasts — each stage validated for temporal consistency and model readiness.

Deep Learning Model Portfolio

Eight architectures evaluated across complexity, accuracy, and interpretability.



Dense & Naive Baseline

Foundational benchmarks and fully-connected networks



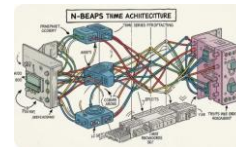
Conv1D

Local temporal pattern extraction via convolutional filters



LSTM

Long short-term memory for sequential dependencies



N-BEATS & Ensemble

Interpretable blocks combined with multivariate ensemble learning

Model Training Infrastructure



TensorFlow Training Stack

→ GPU Acceleration

CUDA-enabled parallel training for rapid iteration

→ Window-Based Learning

Batch sequences generated from sliding windows

→ Chronological Validation

Time-ordered splits prevent future data leakage

→ TensorBoard & Checkpoints

Real-time monitoring with automated model checkpointing

Model Evaluation & Performance



Multi-Metric Evaluation Framework

MAE

Mean Absolute Error — average prediction deviation

RMSE

Root Mean Squared Error — penalizes large errors

MAPE

Mean Absolute Percentage Error — relative accuracy

Prediction Intervals

Uncertainty estimation with confidence bands

Technology Stack



Python

Core language for data engineering, modeling, and evaluation



TensorFlow & Keras

Deep learning framework for model architecture and training



NumPy & Pandas

High-performance data manipulation and numerical computing



GPU & CUDA

Hardware-accelerated training via Google Colab GPU runtime



yFinance & Matplotlib

Market data ingestion and visualization



Enterprise forecasting & planning



BREMIUM AI

Bitcoin
Forecasting Dashboard

Bitcoin AI Forecast

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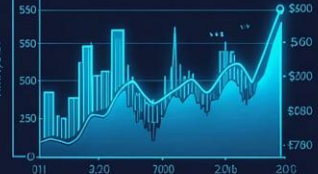


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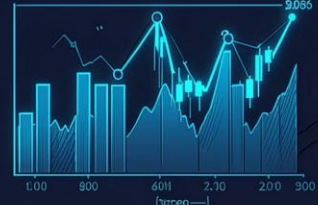
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Microforecasting C2182



Project Summary

End-to-End Pipeline

Raw data to calibrated forecasts

8 Architectures

LSTM, Conv1D, N-BEATS, Ensemble

TensorFlow Engineering

GPU-accelerated, production-ready

Financial AI

Multi-domain forecasting applications

BitPredict demonstrates that deep learning time series forecasting can achieve research-grade accuracy on volatile financial assets — with a reproducible, scalable pipeline built on TensorFlow.

THANK YOU

QUESTIONS?